8 Measuring performance

8.1 No, as $L(\phi) = -\sum_{i=1}^{I} \left(f_{y_i}[x_i, \phi] - \log\Big[\underbrace{\sum_{k'=1}^{K} \exp[f_{k'}[x_i, \phi]]}_{\neq 1}\Big] \right)$

cannot be zero with finite weights and biases
(softmax outputs $]0,1[$)

8.2 $h_1 = 0 + 1x \leadsto \theta_{10} = 0, \ \theta_{11} = 1$
$h_2 = -0.3 + 0.8x \leadsto \theta_{20} = -0.3 \ \ \theta_{21} = 0.8$
$h_3 = -0.65 + 0.7 \leadsto \theta_{30} = -0.65 \ \ \theta_{31} = 0.7$

8.3 $L(\phi) = \sum_{i=1}^{I} (f[x_i, \phi] - y_i)^2$

where $f[x_i, \phi] = \beta + \omega_1 h_1 + \omega_2 h_2 + \omega_3 h_3 = \hat{y}_i$
we can just use the normal equations:

Let $X = \begin{pmatrix} 1 & h_1(x_1) & h_2(x_1) & h_3(x_1) \\ \cdots & \cdots & \cdots & \cdots \\ 1 & h_1(x_I) & h_2(x_I) & h_3(x_I) \end{pmatrix},$

$\beta = \begin{pmatrix} \beta \\ \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix}, \ y = \begin{pmatrix} y_1 \\ \cdots \\ y_I \end{pmatrix}$

Then the closed form solution is given by

$$\hat{\beta} = (X^T X)^{-1} X y$$

8.4 Double descent: testing loss decreases after critical regime

Test performance would decrease (we're now no longer overparameterized), training performance would stay the same (can still memorize entire data set).

8.5 The model will memorize the training data, and because of the log-likelihood, has incentives to model the variance as close to zero as possible. This results in overconfident variance estimates!
⇝ solve by regularizing the variance!

8.6 Let $xy = \sum_{i=1}^{1000} X_i Y_i$ iid $N(0,1)$.

Then

$$\mathbb{E}[xy] = 0 \text{ and } \operatorname{Var}(xy) = 1000,$$

as $\frac{\sqrt{1000}}{\|x\| \, \|y\|} \approx \frac{31.6}{1000} \approx 0.0316$

meaning the angle is close to 90% most of the time, 8.7 $\mathrm{Vol}[r] = \dfrac{r^D \pi^{D/2}}{\Gamma(D/2+1)} \Rightarrow \mathrm{Vol}[0.5] = \dfrac{(1/2)^D \pi^{D/2}}{\Gamma(D/2+1)} = \dfrac{(\pi/4)^{D/2}}{\Gamma(D/2+1)}$

as $(1/4)^{1/2} = 1/2$

Using Stirling's formula:

$$\mathrm{Vol}[0.5] = \frac{(\pi/4)^{D/2}}{\sqrt{\pi D}\left(\frac{D/2}{e}\right)^{D/2}}$$

$$\lim_{D \to \infty} \mathrm{Vol}[0.5] = 0 \qquad \square$$

8.8 For fixed dimension D, $\mathrm{Vol}[r] = c\, r^D$ for some constant c.
The proportion of the volume is

$$P_D = \frac{\mathrm{Vol}[1] - \mathrm{Vol}[0.99]}{\mathrm{Vol}[1]} = 1 - \frac{\mathrm{Vol}[0.99]}{\mathrm{Vol}[1]}$$

$$= 1 - \frac{0.99^D}{1^D} = 1 - 0.99^D$$

$$\lim_{D \to \infty} 1 - 0.99^D = 1 \qquad \square$$